

Second-order polynomial regression model with one independent variable

$$Y_i = b_0 + b_1 X_i + b_2 X_i^2 + e_i$$

Where

- b_0 : the intercept of model
- b_1 : the coefficient of the linear effect (线性效应) of independent variable
- b_2 : the coefficient of the quadratic effect (二次型效应) of independent variable
- X_i : a first-order independent variable, X_i^2 is a second-order independent variable
- e_i is an independent random variable normally distributed.

$$e_i \sim N(0, \sigma^2)$$

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Example of multiple regression analysis in SAS

Monthly Usage	Size of Home
1,182	1,290
1,172	1,350
1,264	1,470
1,493	1,600
1,571	1,710
1,711	1,840
1,804	1,980
1,840	2,230
1,956	2,400
1,954	2,930

Input Power Home;

X1 = Home;

X2 = Home * Home;

Datalines;

```
Proc REG;
Model Power = X1 X2 / All;
Run;
```

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```
Data Dat;
Input Power Home @@;
X1 = Home;
X2 = Home * Home;
Datalines;
1182 1290 1172 1350 1264 1470
1493 1600 1571 1710 1711 1840
1804 1980 1840 2230 1956 2400
1954 2930
;
Proc REG;
Model Power :: X1 X2 / All;
Run;
```

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Model Crossproducts X'X + X'Y + Y'Y

	INTERCEP	X1	X2	Power
INTERCEP	10	18800	37755400	15947
X1	18800	37755400	80939018000	31283250
X2	37755400	80939018000	184303850000000	65306908700
Power	15947	31283250	65306908700	26277083

X'X Inverse, Parameter Estimates, and SSE

	INTERCEP	X1	X2	Power
INTERCEP	26.91557702	-0.02702701	6.36E-06	-1216.143887
X1	-0.027027010000	0.000027591400	-0.000000006580	2.398930177100
X2	0.000006355436	-0.000000006580	0.000000000002	-0.000450040000
Power	1216.143887000000	2.398930177100	-0.000450040000	15332.553629000000

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Source	DF	Sum of Squares	Mean Square	F Value	Prob>F
Model	2	831069.5	415534.8	189.71	0.0001
Error	7	15332.55	2190.365		
C Total	9	846402.1			

Root MSE	46.80133	R-square	0.9819
Dep Mean	1594.7	Adj R-sq	0.9767
C.V.	2.9348		

Variable	DF	Parameter Estimate	Standard Error	T for H0: Parameter=0	Prob> T
INTERCEP	1	-1216.14389	242.8063685	-5.009	0.0016
X1	1	2.39893	0.2458356	9.758	0.0001
X2	1	-0.00045	0.00005908	-7.618	0.0001

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Obs	Dep Var	Predict	Std Err	Lower95%	Upper95%	Lower95%	Upper95%	Residual	Std Err
1	KW	1129.6	30.072	1058.5	1200.7	998	1261.1	52.4359	35.861
2		1172	25.851	1141.1	1263.3	1075.8	1328.6	-30.2136	39.014
3		1264	19.832	1290.9	1338.7	1217.6	1458	-73.7916	42.392
4		1493	17.392	1428.9	1511.2	1352	1588.1	22.9586	43.45
5		1571	17.976	1527.6	1612.6	1451.5	1688.6	8.9359	43.211
6		1711	19.919	1627.1	1721.3	1554	1794.5	36.7685	42.351
7		1804	21.887	1717.6	1821.2	1647.2	1891.6	34.5998	41.368
8		1895	23.348	1848.3	1958.7	1771.8	2019.1	-55.4654	40.561
9		1956	23.611	1893.2	2004.9	1825.1	2073	6.9431	40.409
10		1954	44.734	1843.4	2065	1796.1	2102.3	4.2287	13.755

$y_i = x_i b$

Confidence interval for the average response of a independent variable

Confidence interval for a predicted value of a independent variable

$$S(\hat{b}_0) = \sqrt{MSE x_0^T (X^T X)^{-1} x_0}$$

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Third-order polynomial regression model with one independent variable

$$Y_i = b_0 + b_1 X_i + b_2 X_i^2 + b_3 X_i^3 + e_i$$

The second-order polynomial regression model with two independent variables

$$Y_i = b_0 + b_1 X_{i1} + b_2 X_{i2} + b_3 X_{i1}^2 + b_4 X_{i2}^2 + b_5 X_{i1} X_{i2} + e_i$$

Let's conduct regression analysis of cotton yield (Y) on its three component factors: the boll number per plant (B), the weight per boll (W) and lint (L) as an example



皮棉产量 ← 单株铃数
← 单铃重
← 衣分
产量 = 铃数 × 铃重 × 衣分

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Data of example 4.1.2

i	Boll Number (Boll)	Weight per Boll (g)	Lint (%)	Yield (kg)	$B_i - \bar{B}$	$W_i - \bar{W}$	$L_i - \bar{L}$
1	11.5	4.98	41.39	58.38	-1.39	-0.26	2.36
2	12.05	5.33	40.21	62.5	-0.84	0.09	1.18
3	10.57	5.2	38.68	57.02	-2.32	-0.04	-0.35
4	14.33	5.4	40.27	57.18	1.44	0.16	1.24
5	14.12	6.1	40.28	79.66	1.23	0.86	1.25
6	14.21	4.89	36.84	41.2	1.32	-0.35	-2.19
7	11.76	5.54	39.22	53.3	-1.13	0.30	0.19
8	16.84	6.01	40.91	66	3.95	0.77	1.88
9	10.76	4.84	38.11	50.55	-2.13	-0.40	-0.92
10	13.35	5.11	39.09	53.62	0.46	-0.13	0.06
11	7.01	4.57	36.7	35.16	-5.88	-0.67	-2.33
12	12.25	5.16	37.77	54.76	-0.64	-0.08	-1.26
13	13.2	5.21	39.46	64.72	0.31	-0.03	0.43
14	16.63	5.9	39.57	74.7	3.74	0.66	0.54
15	14.14	5.28	37.82	48.04	1.25	0.04	-1.21
16	13.5	4.36	38.16	39.72	0.61	-0.88	-0.87

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Data Dat;
Input b w l y @@;
bb=b*b;
ww=w*w;
ll=l*l;
bw=b*w;
Cards;

11.5	4.98	41.39	58.38	12.05	5.33	40.21	62.5
10.57	5.2	38.68	57.02	14.33	5.4	40.27	57.18
14.12	6.1	40.28	79.66	14.21	4.89	36.84	41.2
11.76	5.54	39.22	53.3	16.84	6.01	40.91	66
10.76	4.84	38.11	50.55	13.35	5.11	39.09	53.62
7.01	4.57	36.7	35.16	12.25	5.16	37.77	54.76
13.2	5.21	39.46	64.72	16.63	5.9	39.57	74.7
14.14	5.28	37.82	48.04	13.5	4.36	38.16	39.72

Proc Reg;
Model y=b w bb ww bw /noint;
Run;

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Analyze the linear relationship between the cotton yield (Y) and the boll number per plant (B) and the weight per boll (W) in the second-order polynomial regression model through the origin

$$Y_i = b_B B_i + b_W W_i + b_{BB} B_i^2 + b_{WW} W_i^2 + b_{BW} B_i W_i + e_i$$

where $BB = B \times B$, $WW = W \times W$, $BW = B \times W$.

The coefficient of multiple determination $R^2 = SSR/SSTO = 0.9920$

The adjusted coefficient of multiple determination $R_a^2 = 1 - MSE/MSTO = 0.9884$

Multiple Linear Regression

$$Y_i = b_B B_i + b_W W_i + b_L L_i + e_i$$

The coefficient of multiple determination $R^2 = SSR/SSTO = 0.9856$

The adjusted coefficient of multiple determination $R_a^2 = 1 - MSE/MSTO = 0.9822$

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The analysis result in the second-order polynomial regression model through the origin for the cotton yield

Variable	Estimate	Standard Error	t-value	P value
B	-0.82	8.99	-0.03	0.929
W	5	24.28	0.21	0.841
BB	-0.74	0.45	-1.66	0.128
WW	-3.07	6.57	-0.47	0.649
BW	3.71	3.43	1.08	0.303

The results indicates that the effects of the second-order variable (BB), (WW) and (BW) do not reach the significance level of 5%

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The deviation from the mean of independent variable X is usually adopted in the analysis of polynomial regression model.

The second-order polynomial regression model with one variable

$$Y_i = b_0 + b_1 (X_i - \bar{X}) + b_2 (X_i - \bar{X})^2 + e_i$$

$$= b_0 + b_1 x_i + b_2 x_i^2 + e_i$$

Where x_i is the deviation from average of the independent variables

The third-order polynomial regression model with one variable

$$Y_i = b_0 + b_1 (X_i - \bar{X}) + b_2 (X_i - \bar{X})^2 + b_3 (X_i - \bar{X})^3 + e_i$$

$$= b_0 + b_1 x_i + b_2 x_i^2 + b_3 x_i^3 + e_i$$

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The second-order polynomial regression model with two variables

$$Y_i = b_0 + b_1(X_{i1} - \bar{X}_1) + b_2(X_{i2} - \bar{X}_2) + b_3(X_{i1} - \bar{X}_1)^2 + b_4(X_{i2} - \bar{X}_2)^2 + b_5(X_{i1} - \bar{X}_1)(X_{i2} - \bar{X}_2) + e_i$$

$$= b_0 + b_1x_{i1} + b_2x_{i2} + b_3x_{i1}^2 + b_4x_{i2}^2 + b_5x_{i1}x_{i2} + e_i$$

This model was used to analyze the linear relationship between the yield of cotton (Y) and the boll number per plant (B) and the weight per boll (W) in example 4.1. The results were presented in following table

Data Dat;
Input b w l y @@;
bb=b*b;
ww=w*w;
ll=l*l;
bw=b*w;
Cards;

```
-1.39 -0.26 2.36 58.38 -0.84 0.09 1.18 62.5
-2.32 -0.04 -0.35 57.02 1.44 0.16 1.24 57.18
1.23 0.86 1.25 79.66 1.32 -0.35 -2.19 41.2
-1.13 0.30 0.19 53.3 3.95 0.77 1.88 66
-2.13 -0.40 -0.92 50.55 0.46 -0.13 0.06 53.62
-5.88 -0.67 -2.33 35.16 -0.64 -0.08 -1.26 54.76
0.31 -0.03 0.43 64.72 3.74 0.66 0.54 74.7
1.25 0.04 -1.21 48.04 0.61 -0.88 -0.87 39.72
```

Proc Reg;
Model y=b w bb ww bw /noint;
Run;

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Table 5.9: ANOVA Table in the second-order polynomial regression model through the origin using deviation of independent variables for the cotton yield

Source	DF	SS	MS	F-value	P-value
Regression	5	25239	5047.77	2.04	0.15
Residual	11	27156	2468.73		
Total	16	52395			

The coefficient of multiple determination

$$R^2 = SSR/SSTO = 0.4817$$

The adjusted coefficient of multiple determination

$$R_a^2 = 1 - MSE/MSTO = 0.2461$$

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Table 5.10: Estimates of parameters based on the second-order polynomial regression model through the origin using deviation of independent variables for the cotton yield

Variable	Estimate	Standard Error	t value	P value
B	0.449	8.089	0.06	0.9568
W	43.964	43.489	1.01	0.3338
BB	5.884	5.341	1.1	0.2341
WW	94.827	50.238	1.89	0.0857
BW	-33.009	38.639	-0.98	0.3464

Large difference could be observed between the results (ANOVA, estimate of parameter) of the second-order polynomial models using original values and deviations from average of independent variables.

? Comments or indications on the difference

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§ 5.2 Regression Model with Indicator Variables

Independent variables in regression model:

- Quantitative Variables (数量变量)
- Qualitative Variables (质量变量).

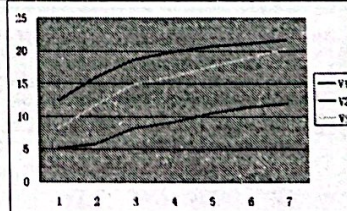
Indicator Variable (指示变量): which take scalar values of 0 or 1 to denote the difference level of qualitative variable by transformation.

Indicator variables are sometimes called dummy variables (假变量).

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3个棉花品种不同时期的单株成铃数

Date	V1	V2	V3
35	12.29	5.05	8.06
40	15.86	5.78	11.63
45	18.57	8.15	14.87
50	19.75	9.11	15.97
55	20.61	10.45	17.52
60	21.03	11.32	18.82
65	21.6	11.9	20.31

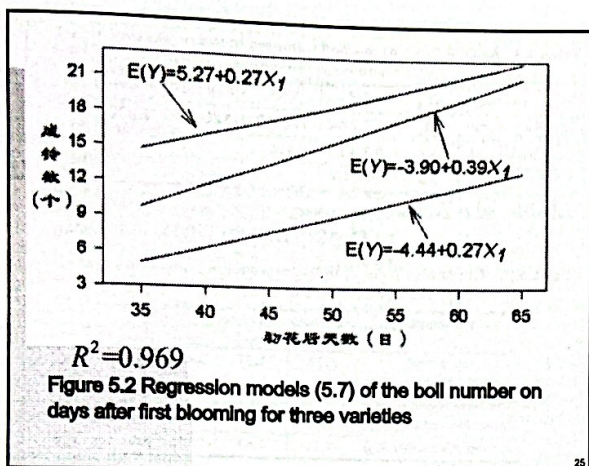


Variable	DF	Estimate	Error	t Value	Pr> t
V1: $\hat{Y} = -3.90 + 0.384X$, $r^2 = 0.95$					
Intercept	1	-3.89571	2.84729	-1.90	0.1154
Day	1	0.38414	0.04815	9.57	<.0001
V2: $\hat{Y} = -3.30 + 0.24X$, $r^2 = 0.97$					
Intercept	1	-3.39500	0.96933	-3.40	0.0193
Day	1	0.24256	0.01901	12.75	<.0001
V3: $\hat{Y} = 4.13 + 0.288X$, $r^2 = 0.859$					
Intercept	1	4.13357	2.66813	1.55	0.1309
Day	1	0.28793	0.05217	5.52	0.0027

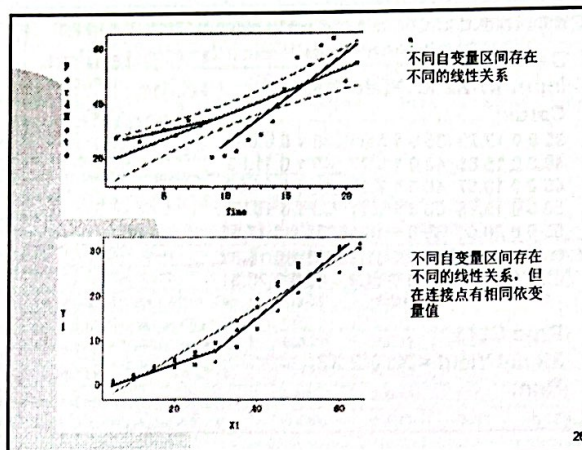
- 不同品种成铃数与生育期线性关系不尽相同, 如何分析数据?
- 可采用哪些分析方法? 优缺点?

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不同 regression model 之间
不能比较



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用同一个模型解释3个品种，比较3个品种的截距和斜率差异。

If the dependent variable Y_i is determined by a quantitative variable and a qualitative variable, and the qualitative variable has c categories ($c-1$ independent variables), then the relationship between Y_i and independent variables can be expressed by the following regression model ($i = 1, 2, \dots, n$),

$$Y_i = b_0 + b_1X_{i1} + \sum_{j=2}^c b_jX_{ij} + \varepsilon_i$$

where

- b_0 is the regression intercept
- b_1 is the coefficient of the quantitative independent variable
- b_j is the coefficient of the indicator variable X_{ij}
- ε_i is an independent random variable subjected to normal distribution $\varepsilon_i \sim N(0, \sigma^2)$

Using this term to adjust the intercept of the model

This regression model with indicator variables is based on the hypothesis: different category of regression models have identical slope.

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Example A cotton experiment with 3 varieties was conducted. The average boll numbers per plant (dependent variable Y) and the number of days after the first blooming (July 1st) (quantitative variable X_1) were investigated at different growth stages. The variety is qualitative variable with 3 categories ($c = 1, 2, 3$) which can be scaled by two indicator variables X_2 and X_3 .

Variety 1: $X_2 = 1, X_3 = 0$;
 Variety 2: $X_2 = 0, X_3 = 1$;
 Variety 3: $X_2 = 0, X_3 = 0$;

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The regression model of dependent variable Y_i on independent variables can be expressed in following equation ($i = 1, 2, \dots, 21$)

$$Y_i = b_0 + b_1X_{i1} + b_2X_{i2} + b_3X_{i3} + \varepsilon_i$$

This model describe the linear relationship between the average boll number per plant of three varieties and the number of days after the first blooming of three varieties

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The linear regression models for three varieties have the same slope but different intercepts.

The expectation $E(Y)$ of the dependent variable varies among varieties.

For variety 1, $E(Y) = b_0 + b_1X_1 + b_2(1) + b_3(0) = (b_0 + b_2) + b_1X_1$

For variety 2, $E(Y) = b_0 + b_1X_1 + b_2(0) + b_3(1) = (b_0 + b_3) + b_1X_1$

For variety 3, $E(Y) = b_0 + b_1X_1 + b_2(0) + b_3(0) = b_0 + b_1X_1$

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Data day;
Input X1 X2 X3 Yield @@;
Cards;
35 0 0 12.29 35 0 1 5.05 35 1 0 8.06
40 0 0 15.86 40 0 1 5.78 40 1 0 11.63
45 0 0 18.57 45 0 1 8.15 45 1 0 14.87
50 0 0 19.75 50 0 1 9.11 50 1 0 15.97
55 0 0 20.61 55 0 1 10.45 55 1 0 17.52
60 0 0 21.03 60 0 1 11.32 60 1 0 18.82
65 0 0 21.6 65 0 1 11.9 65 1 0 20.31
;
Proc GLM;
Model Yield = X1 X2 X3;
Run;

Pay attention to the arrangement and input of data

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Table 5.1 ANOVA table of the boll numbers in the regression model with same slope for three cotton varieties.

Source	DF	SS	MS	F value	P value
Regression	3	537.384	179.13	128.12	0.0001
Residual	17	23.77	1.4		
Total	20	561.15			

The coefficient of multiple determination
 $R^2 = SSR/SSTO = 537.720/561.478 = 0.958$

Table 5.2 The analysis result of the regression parameters for the boll numbers of cotton

Parameter	Estimate	Standard Error	t value	P value
b_0 Intercept	3.29	1.37	2.41	0.03
b_1 Number of Days after The First Blooming (X_1)	0.30	0.03	11.81	0.00
b_2 Indicator Variable (K_2)	-3.22	0.63	-5.09	0.00
b_3 Indicator Variable (K_3)	-9.71	0.63	-15.36	0.00

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The boll numbers per plant can be predicted by the number of days using following 3 models for different varieties, respectively.

Variety 1: $\hat{Y} = (\hat{b}_0 + \hat{b}_2) + \hat{b}_1 X_1 = 0.07 + 0.30 X_1$
 Variety 2: $\hat{Y} = (\hat{b}_0 + \hat{b}_3) + \hat{b}_1 X_1 = -6.42 + 0.30 X_1$
 Variety 3: $\hat{Y} = \hat{b}_0 + \hat{b}_1 X_1 = 3.29 + 0.30 X_1$

Fig 5.1 The regression model of the boll numbers of three varieties on the number of days after the first blooming for three varieties

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If the intercepts and slopes in the different category of regression models of the dependent variable Y on the independent variables are all different, then the relationship between Y_i and the independent variables can be expressed by the following equation ($i=1,2,\dots,n$)

$$Y_i = b_0 + b_1 X_{i1} + \sum_{j=2}^c b_j X_{ij} + \sum_{j=c-1+j}^c b_{c-1+j} X_{i1} X_{ij} + \varepsilon_i$$

调整截距 (Adjusting Intercept) 调整斜率 (Adjusting Slope) 斜率

b_0 is the regression intercept
 b_1 is the coefficient of the quantitative independent variable
 $b_j, b_{(c-1+j)}$ is the coefficient of the indicator variables
 ε_i is an independent random variable following normal distribution
 $\varepsilon_i \sim N(0, \sigma^2)$

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Fig 5.2 The regression model (5.7) the boll numbers on the number of days after the first blooming for three varieties

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Now the analysis method in regression model with indicator variables will be introduced in detail, using the cotton experimental data in example 5.1 as an example.

The regression model of dependent variable Y_i on independent variables can be expressed in following equation ($i=1,2,\dots,21$),

$$Y_i = b_0 + b_1 X_{i1} + b_2 X_{i2} + b_3 X_{i3} + b_4 X_{i1} X_{i2} + b_5 X_{i1} X_{i3} + \varepsilon_i$$

Variety 1 $E(Y) = b_0 + b_1 X_1 + b_2 \times 1 + b_3 \times 0 + b_4 X_1 \times 1 + b_5 X_1 \times 0$
 $= (b_0 + b_2) + (b_1 + b_4) X_1$
 Variety 2 $E(Y) = b_0 + b_1 X_1 + b_2 \times 0 + b_3 \times 1 + b_4 X_1 \times 0 + b_5 X_1 \times 1$
 $= (b_0 + b_3) + (b_1 + b_5) X_1$
 Variety 3 $E(Y) = b_0 + b_1 X_1 + b_2 \times 0 + b_3 \times 0 + b_4 X_1 \times 0 + b_5 X_1 \times 0$
 $= b_0 + b_1 X_1$

There are different regression intercepts and slopes for above three models.

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$b_4 = b_5 = 0$. 斜率相同.
 $b_2 = b_3 = 0$. 截距相同.

Data d;
Input X1 X2 X3 Yield @@;
X12= X1*X2;
X13= X1*X3;
Cards;

35 0 0 12.29 35 0 1 5.05 35 1 0 8.06
40 0 0 15.86 40 0 1 5.78 40 1 0 11.63
45 0 0 18.57 45 0 1 8.15 45 1 0 14.87
50 0 0 19.75 50 0 1 9.11 50 1 0 15.97
55 0 0 20.61 55 0 1 10.45 55 1 0 17.52
60 0 0 21.03 60 0 1 11.32 60 1 0 18.82
65 0 0 21.6 65 0 1 11.9 65 1 0 20.31

; Proc GLM;
Model Yield = X1 X2 X3 X12 X13;
Run;

Pay attention
to the
arrangement
and input of
data chunks

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Table 5.3 ANOVA table of the boll numbers of cotton in the regression model with different slopes

Source	DF	SS	MS	F value	P value
Regression	5	544.72	108.94	99.44	0.0001
Residual	15	16.43	1.11		
Total	20	561.15			

Table 5.4 Result of the statistical analysis of the regression parameters for the boll number of cotton

Variable	Estimate	SE	t value	P value
Intercept	4.134	2.017	2.05	0.058
Number of Days after The First Blooming (X_1)	0.288	0.04	7.28	0
Indicator Variable (X_2)	-8.029	2.853	-2.81	0.013
Indicator Variable (X_3)	-7.429	2.853	-2.6	0.02
Indicator Variable (X_1X_2)	0.096	0.065	1.72	0.106
Indicator Variable (X_1X_3)	-0.046	0.056	-0.81	0.428

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The coefficient of multiple determination

$$R^2 = SSR/SSTO = 544.72/561.15 = 0.971$$

Since the p-value of the indicator variable X_1X_3 is the largest, thus, this variable is deleted, analysis is performed again based on the following model ($i=1,2, \dots, 21$),

$$Y_i = b_0 + b_1X_{i1} + b_2X_{i2} + b_3X_{i3} + b_4X_{i1}X_{i2} + e_i$$

Variety 1 $E(Y) = b_0 + b_1X_1 + b_2(1) + b_3(0) + b_4X_1(1)$
 $= (b_0 + b_2) + (b_1 + b_4)X_1$

Variety 2 $E(Y) = b_0 + b_1X_1 + b_2(0) + b_3(1) + b_4X_1(0)$
 $= (b_0 + b_3) + b_1X_1$

Variety 3 $E(Y) = b_0 + b_1X_1 + b_2(0) + b_3(0) + b_4X_1(0)$
 $= b_0 + b_1X_1$

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Table 5.5 ANOVA table based on the regression model with indicator variables for the boll number of cotton

Source	DF	SS	MS	F value	P value
Regression	4	543.99	136	126.81	0.0001
Residual	16	17.16	1.07		
Total	20	561.15			

The coefficient of multiple determination is

$$R^2 = SSR/SSTO = 543.99/561.15 = 0.969$$

Table 5.6 Result of statistical analysis of the regression parameters for the boll number of cotton

Variable	Estimate	SE	t value	P value
Intercept	5.27	1.44	3.67	0.002
Number of Days after The First Blooming (X_1)	0.27	0.03	9.58	0.000
Indicator Variable (X_2)	-9.17	2.46	-3.73	0.002
Indicator Variable (X_3)	-9.71	0.55	-17.54	0.000
Indicator Variable (X_1X_2)	0.12	0.85	2.48	0.024

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Using the following models for different varieties, the boll numbers per plant can be predicted by the number of days after the first blooming

For variety 1: $\hat{Y} = (\hat{b}_0 + \hat{b}_2) + (\hat{b}_1 + \hat{b}_4)X_1 = -3.90 + 0.39X_1$

For variety 2: $\hat{Y} = (\hat{b}_0 + \hat{b}_3) + \hat{b}_1X_1 = -4.44 + 0.27X_1$

For variety 3: $\hat{Y} = \hat{b}_0 + \hat{b}_1X_1 = 5.27 + 0.27X_1$

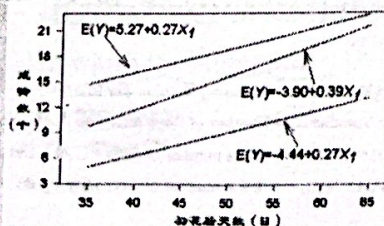
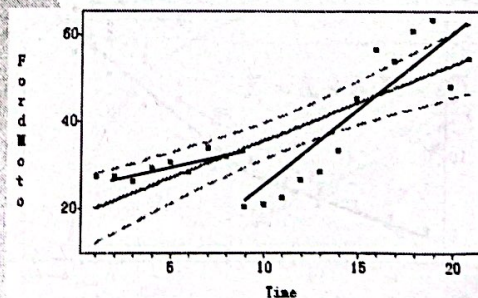


Fig The regression model of the boll numbers on the number of days after the first blooming for three varieties

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If the linear relationships between Y and X may vary at diverse ranges of X , then the piecewise regression model (分段回归模型) can be employed.



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Date	Season	Time	FordMoto	DeuMark	S&P500
1994	3	1	27.75	1.55	466.96
1994	4	2	27.88	1.55	455.19
1995	1	3	26.88	1.38	493.15
1995	2	4	29.75	1.38	539.35
1995	3	5	31.12	1.42	578.77
1995	4	6	28.88	1.43	614.57
1996	1	7	34.38	1.48	647.07
1996	2	8	32.38	1.52	668.5
1996	3	9	20.7	1.53	687.33
1996	4	10	21.28	1.55	740.74
1997	1	11	23.02	1.68	757.12
1997	2	12	27.07	1.74	885.14
1997	3	13	28.94	1.77	947.28
1997	4	14	33.78	1.79	970.43
1998	1	15	45.81	1.85	1101.75
1998	2	16	57	1.81	1133.84
1998	3	17	54.25	1.68	1017.01
1998	4	18	61.44	1.67	1229.23
1999	1	19	63.94	0.93	1286.37
1999	2	20	-8.5	0.97	1372.71
1999	3	21	54.88	0.94	1282.71

- The quantitative variable (X_1) is time (1~21)
- The indicator Variable (X_2): 0 before time 8 (includes), 1 afterwards

$$Y_i = b_0 + b_1 X_{i1} + b_2 X_{i2} + b_3 X_{i1} X_{i2} + e_i$$

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Full Model $Y = b_0 + b_1 X_1 + b_2 X_2 + b_3 X_1 X_2 + e$

For the Ford stock in the first 8 seasons

$$E(Y) = b_0 + b_1 X_1 + b_2(0) + b_3 X_1(0)$$

$$= b_0 + b_1 X_1 = 26.00 + 0.86 X_1$$

Root E	6.04075	R-Square	0.8483
Dependent mean	37.12524	Adj R-Sq	0.8121

For the Ford stock in late 13 seasons

Parameter	Estimate	Error	t-value	Pr> t
b0	26.00536	4.70692	5.52	<.0001
b1	0.86048	0.93211	0.92	0.3688
b2	-40.61310	8.37183	-4.85	0.0001
b3	2.88573	1.83408	2.79	0.0126

$$E(Y) = b_0 + b_1 X_1 + b_2(1) + b_3 X_1(1)$$

$$= (b_0 + b_2) + (b_1 + b_3) X_1$$

$$= -14.60 + 3.75 X_1$$

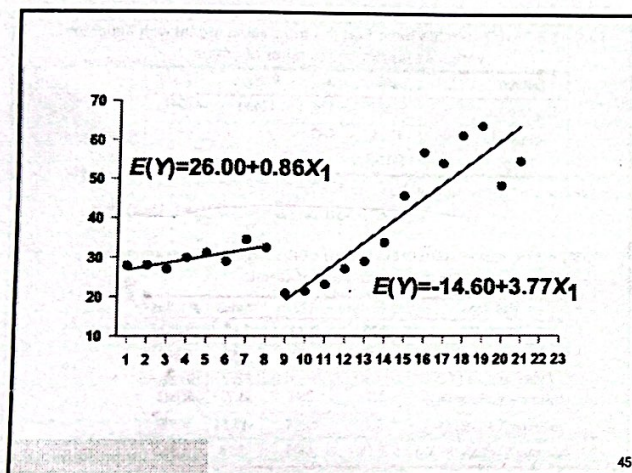
Simple Linear Regression

$$\hat{Y} = 18.17 + 1.76 X$$

$$R^2 = 0.589$$

Note: 直接采用简单线性回归模型分析, 决定系数比采用分段线性模型分析结果小得多

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前面分析发现 b_1 不显著, 移除该变量重新分析

Reduced Model $Y = b_0 + b_2 X_2 + b_3 X_1 X_2 + e$

For the Ford stock in the first 8 seasons

$$E(Y) = b_0 + b_2(0) + b_3 X_1(0)$$

$$= b_0 = 29.88$$

For the Ford stock after 8 seasons,

$$E(Y) = b_0 + b_2(1) + b_3 X_1(1)$$

$$= (b_0 + b_2) + b_3 X_1$$

$$= -14.61 + 3.75 X_1$$

Simple Linear Regression

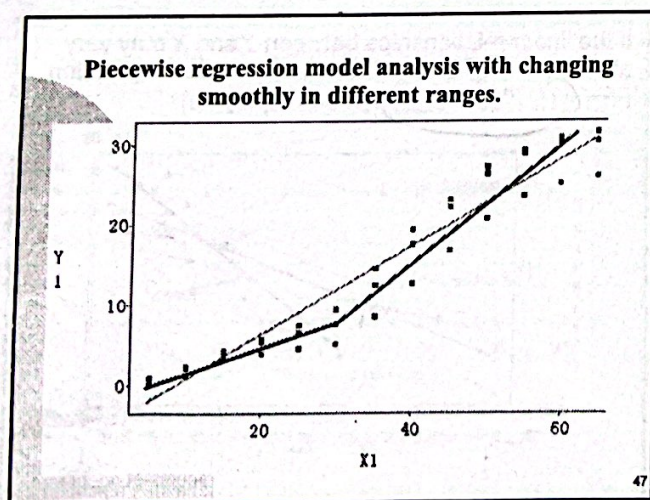
$$\hat{Y} = 18.17 + 1.76 X$$

$$r^2 = 0.589$$

Root MSE	6.01590	R-Square	0.8323
Dependent Mean	37.12524	Adj R-Sq	0.8137

Variable	DF	Estimate	Error	t-Value	Pr> t
Intercept	1	29.87750	2.12894	14.05	<.0001
X2	1	-44.48525	7.21454	-6.17	<.0001
X3	1	3.74621	0.44593	8.40	<.0001

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Example: The average flowering number per plant in diverse growth stages was investigated in cotton trial. The flowering behavior of cotton in the first 30 days (before autumn begins) is found different from that in the following 35 days (after the autumn begins), indicating that regression model of the average flowering number per plant on the number of days after the first blooming is probably different in the intercept and slope.

Dependent Variable Y = Flowering Number per Plant

Qualitative Variable X_1 = Number of Days after The First Blooming (July 1st)

Indicator Variable $X_{12} = 0$ If the number of days after the first blooming ≤ 30

Indicator Variable $X_{12} = 1$ If the number of days after the first blooming > 30

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Since the flowering number is 0 when the number of days is 0, the relationship between Y_i and the independent variables can be described in the regression model through the origin,

$$Y_i = b_1 X_{i1} + b_2 (X_{i1} - 30) X_{i2} + e_i$$

Regression model with indicator variables (5.8) actually describes the different linear relationships between the average flowering number per plant and the number of days after the first blooming in two periods.

强制过原点; 且两段相接

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For the cotton in 30 days before autumn begins

$$E(Y) = b_1 X_1 + b_2 (X_1 - 30)(0) = b_1 X_1$$

For the cotton in the 35 days after autumn begins

$$E(Y) = b_1 X_1 + b_2 (X_1 - 30)(1) = -30b_2 + (b_1 + b_2) X_1$$

Table 5.8 The results of statistical analysis on the regression parameters for the flowering number per plant of cotton

Variable	Estimator	SE	t* value	P value
Number of Days after The First Blooming (X_1)	0.294	0.025	11.613	0.0001
Indicator Variable ($(X_1 - 30)X_2$)	0.393	0.061	6.418	0.0001

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Table 5.7 ANOVA table of the boll number of cotton based on the regression model with indicator variables

Source	DF	SS	MS	F value	P value
Regression	2	12278.72	6139.36	902.499	0.0001
Residual	37	251.697	6.803		
Total	39	12530.417			

The coefficient of multiple determination

$$R^2 = SSR/SSTO = 12278.72/12530.417 = 0.980$$

Nearly above 98% of the variation of the flowering number is determined by these two independent variables.

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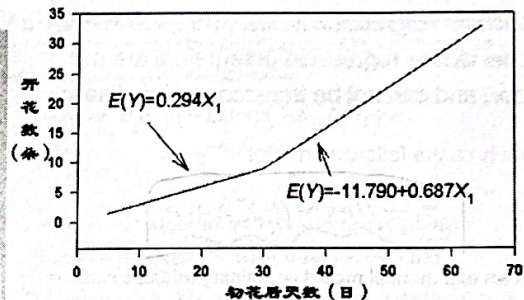


Figure Regression model of the flowering number per plant on the number of days after the first blooming for one variety

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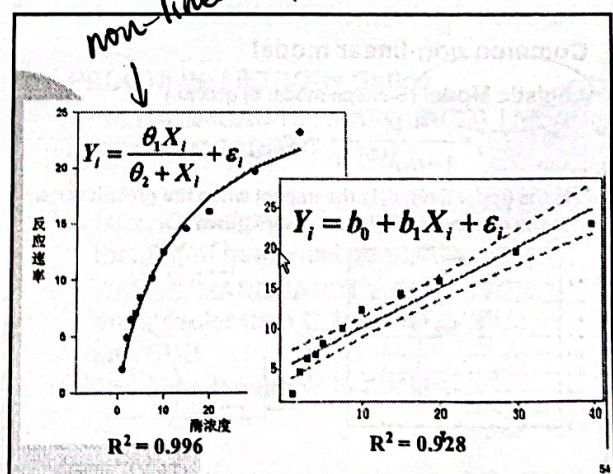
§ 5.3 Intrinsically Linear and Non-Linear Regression

线性回归模型(linear regression model):

A model in which the regression parameters are linear

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First-order linear model

$$Y_i = b_0 + b_1 X_{i1} + b_2 X_{i2} + \dots + b_p X_{ip} + e_i$$

Second-order linear model

$$Y_i = b_0 + b_1 X_{i1} + b_2 X_{i2} + b_3 X_{i1}^2 + b_4 X_{i2}^2 + b_5 X_{i1} X_{i2} + e_i$$

Linear model with the dependent and independent variables transformed

$$\log Y_i = b_0 + b_1 \sqrt{X_{i1}} + b_2 \exp(X_{i2}) + e_i$$

Intrinsically Linear Regression Model (内蕴线性回归模型):

Although the regression parameters of some models are not linear, but they can transformed into linear parameters



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Let's take logarithm on the two sides of this model

$$Y_i = \theta_0 [\exp(\theta_1 X_i)] e_i$$

then, it can be changed to linear regression model

$$\log(Y_i) = \ln(\theta_0) + \theta_1 X_i + \ln(e_i)$$

Suppose

$$Y'_i = \log(Y_i) \quad b_0 = \log(\theta_0), \quad b_1 = \theta_1 \quad e'_i = \ln(e_i)$$

at last, the model above can be written as

$$Y'_i = b_0 + b_1 X_{i1} + e'_i$$

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Nonlinear Regression Model (非线性回归模型): a model whose regression parameters are not linear, and can not be transformed into linear.

such as the following model

$$Y_i = \theta \exp(\theta_2 X_i) + e_i$$

is an exponential model containing additive random error items, so it can not be transformed into linear regression model.

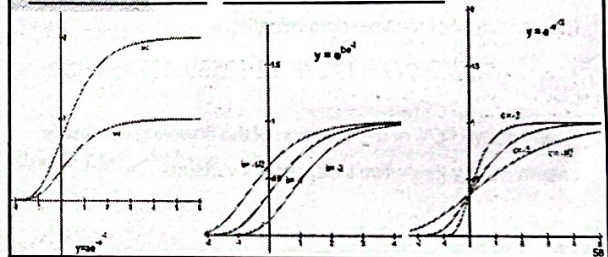
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Common non-linear model

Gompertz Model (S-shape model of growth)

$$Y_i = \theta_1 e^{-\theta_2} e^{-\theta_3 X_i} + \varepsilon_i$$

θ_1 is the upper limit, $-\theta_3$ is the rate of growth, $e = 2.71828...$



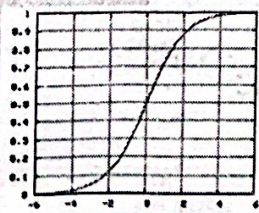
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Common non-linear model

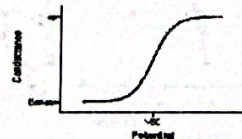
Logistic Model (S-shape model of growth)

$$Y_i = \frac{\theta_1}{1 + \theta_2 e^{-\theta_3 X_i}} + \varepsilon_i$$

θ_1 is the upper limit; θ_2 is the impact when the growth is near to the upper limit, θ_3 is the rate of growth



$$Y = \text{Bottom} + \frac{(\text{Top} - \text{Bottom})}{1 + \exp\left(\frac{V60 - X}{\text{Slope}}\right)}$$



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Unadvisable Expressions of Curve Regression (p=213)

表 11.1 常用曲线回归方程的直线化方法

曲线回归方程	经尺度转换的新变量和新参数				转换后的直线回归方程
$y = ae^{bx} + c$	y'	x'	a'	b'	
$y = ae^{bx} \quad (a > 0)$	$y' = \ln y$		$a' = \ln a$		$y' = a' + bx$
$y = ae^{bx} \quad (a > 0)$	$y' = \ln y$		$a' = \ln a$		$y' = a' + bx$
$y = a + b \ln x \quad (a > 0)$		$x' = \ln x$			$y' = a' + bx'$
$y = ae^{bx} \quad (a > 0, b > 0)$	$y' = \ln y$	$x' = \ln x$	$a' = \ln a$		$y' = a' + bx'$
$y = \frac{x}{a + bx} \quad (a \neq 0, b \neq 0)$	$y' = \frac{y}{x}$				$y' = a' + bx'$
$y = \frac{a + bx}{c} \quad (z \neq 0)$	$y' = y$				$y' = a' + bx'$
$y = \frac{1}{a + bx} \quad (z \neq 0, b \neq 0)$	$y' = \frac{1}{y}$				$y' = a' + bx'$
$y = \frac{1}{1 + ae^{-bx}} \quad (a > 0)$	$y' = \ln\left(\frac{1-y}{y}\right)$		$a' = \ln a$		$y' = a' - bx'$

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$$\hat{y} = \hat{b}_0 + \hat{b}_1 x_i$$

$$y = b_0 + b_1 x_i + \varepsilon_i$$

Estimating The Parameters of Non-Linear Regression General Expression of Non-Linear Regression Model

$$y_i = f(x_i, \theta) + \varepsilon_i \sim N(f(x_i, \theta), \sigma^2) \quad i = 1, \dots, n$$

$$x_i = (x_{i1} \dots x_{iq})^T \quad \text{每个个体调查了 } q \text{ 个属性指标}$$

$$\theta = (\theta_1, \dots, \theta_p)^T \quad p \text{ 个参数组成的参数向量}$$

Non-Linear Regression Model in Matrix Form

$$y = f(X, \theta) + e$$

$$y = (y_1 \dots y_n)^T \quad X = \begin{pmatrix} x_{11} & \dots & x_{1q} \\ \vdots & & \vdots \\ x_{n1} & \dots & x_{nq} \end{pmatrix}$$

$$f(X, \theta) = (f(x_1, \theta) \dots f(x_n, \theta))^T = f(\theta)$$

$$e = y - f(\theta), \hat{e} = y - f(\hat{\theta})$$

$$MSE = SSE(\hat{\theta}) / (n - p)$$

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Estimating The Parameters of Non-Linear Regression General Expression of Non-Linear Regression Model

$$Q = (y - f(X, \theta))^T (y - f(X, \theta)) = \sum_{i=1}^n (y_i - f(x_i, \theta))^2$$

We can get the Jacobian matrix of $f(\theta)$ by conducting the partial derivative on the vector of regression parameters in all functions $f(\theta)$

$$Z(\theta) = \frac{\partial f(X, \theta)}{\partial \theta^T} = \begin{pmatrix} \partial f(x_1, \theta) / \partial \theta_1 & \dots & \partial f(x_1, \theta) / \partial \theta_p \\ \vdots & & \vdots \\ \partial f(x_i, \theta) / \partial \theta_1 & \dots & \partial f(x_i, \theta) / \partial \theta_p \\ \vdots & & \vdots \\ \partial f(x_n, \theta) / \partial \theta_1 & \dots & \partial f(x_n, \theta) / \partial \theta_p \end{pmatrix}$$

$$\text{If } f(X, \theta) = X\theta, \text{ then } Z(\theta) = X$$

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The estimator of the regression parameter $\hat{\theta}$ by Gauss-Newton iteration is an approximation to the solution of the least of squares. If the iteration is convergent, then the $\hat{\theta}$ can be formulated in following Taylor series with regression parameter θ

$$\hat{\theta} = \theta + (Z^T(\theta)Z(\theta))^{-1}Z^T(\theta)(y - f(X, \theta))$$

If $f(X, \theta) = X\theta$, then $Z(\theta) = X$

$$\hat{\theta} = \hat{\theta} + (X^T X)^{-1} X^T (y - X\hat{\theta}) \quad \hat{\theta} = (X^T X)^{-1} X^T y$$

注：上面估计式可知，多变量线性模型是非线性模型的一种特殊情况

If the experimental sample is large enough, then $\hat{\theta}$ follows an asymptotic multivariate normal distribution

$$\hat{\theta} \sim MVN(\theta, \sigma^2(Z^T(\theta)Z(\theta))^{-1})$$

多变量线性模型

$$\hat{\theta} = (X^T X)^{-1} X^T y \sim MVN(b, \sigma^2(X^T X)^{-1})$$

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Example: Suppose there's a non-linear relationship between the boll number of cotton per plant in different periods (Y) and the number of days after the first blooming (X), which can be fitted by a growth model (Gompertz model)

$$Y_i = \theta_1 \exp[-\theta_2 \exp(-\theta_3 X_i)] + e_i$$

$$Y_i = \theta_1 e^{-\theta_2 e^{-\theta_3 X_i}} + e_i$$

The observed values of boll numbers per plant from Jul. 5th to Sep. 3rd with interval of 5 days

$$y^T = [0.75 \ 2.0 \ 4.0 \ 4.75 \ 5.25 \ 5.5 \ 7.75 \ 10.13 \ 12.26 \ 13.14 \ 13.52 \ 14.15 \ 14.53]$$

The numbers of days after the first blooming

$$X^T = [5 \ 10 \ 15 \ 20 \ 25 \ 30 \ 35 \ 40 \ 45 \ 50 \ 55 \ 60 \ 65]$$

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SAS program

```
Data Cotton;
Input Day Boll @@;
Datalines;
5 0.75 10 2.0 15 4.0
20 4.75 25 5.25 30 5.5
35 7.75 40 10.13
45 12.26 50 13.14
55 13.52 60 14.15
65 14.53
;
PROC NLIN METHOD=GAUSS;
PARMS A=20.0 B=1.0 C=0.0;
MODEL Boll= A*EXP(-B*EXP(-C*Day));
RUN;
```

Numerical Computing Method

Initial Value of Iteration

Model

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- PROC NLIN: METHOD= Option
 - METHOD=GAUSS | MARQUARDT | NEWTON | GRADIENT | DUD
 - METHOD= Option specify the method of numerical computing
 - GAUSS, MARQUARDT and NEWTON are steadier than GRADIENT(梯度法) and DUD
 - The default method is GAUSS

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NLIN 模块

The Gauss-Newton method was used to estimate the non-linear regression parameters, the initial value is $\theta_1=20.0$, $\theta_2=1.0$, $\theta_3=0$. the estimation of parameters could be achieved only 8 times iteration.

Iteration Time	θ_1	θ_2	θ_3	Residual Sum of Squares
0	20.000000	1.000000	0.000000	299.195285
1	20.000000	2.051595	0.033654	63.133251
2	20.432037	2.946603	0.036006	9.581773
3	18.777912	3.386069	0.041683	6.858041
4	19.060303	3.478358	0.041680	6.763691
5	19.021486	3.487013	0.041838	6.763439
6	19.028501	3.486338	0.041816	6.763435
7	19.027362	3.486476	0.041820	6.763435
8	19.027552	3.486454	0.041819	6.763435

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Table of The Analysis of Variance

Source	DF	SS	MS
Regression	3	1173.9544654	391.3181551
Residual	10	6.7634346	0.6763435
Total	13	1180.7179000	

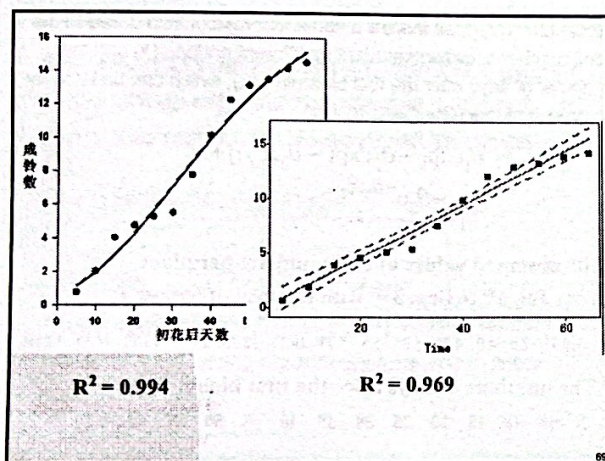
$$R^2 = 0.994272$$

Parameter	Estimate	Asymptotic Random Error	Lower Bound of 95% Confidential Interval	Upper Bound of 95% Confidential Interval
θ_1	19.028	2.521	13.411	24.645
θ_2	3.480	0.449	2.487	4.486
θ_3	0.042	0.008	0.023	0.061

All the regression parameters reach the significance level of 5%, so the boll number of cotton per plant can be predicted by the number of days after the first blooming using the following non-linear regression model

$$\hat{Y} = 19.028 \exp[-3.486 \exp(-0.042X)]$$

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Example 2

In study of enzyme dynamics, there is a non-linear relationship between the reaction rate (Y) and the density of the enzyme

$$Y_i = \frac{\theta_1 X_i}{\theta_2 + X_i} + \varepsilon_i$$

The 11 observed values are as follows

X_i	1	2	3	4	5	8	10	15	20	30	40
Y_i	2.1	4.9	6.5	7.0	8.4	10.2	12.5	14.6	16.1	19.7	23.2

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The results based on the non-linear regression

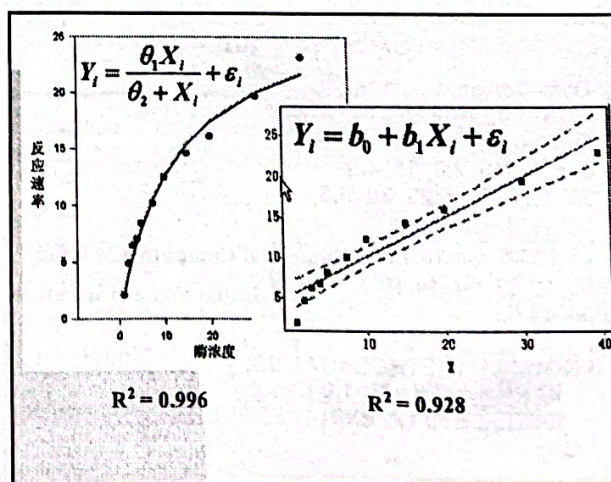
Table of the analysis of variance

Source	DF	SS	MS
Regression	2	1841.7443917	920.8721959
Residual	9	7.4756083	0.8306231
Total	11	1849.2200000	

$$R^2 = 0.995957$$

Parameter	Estimator	Asymptotic Random Error	Lower Bound of The 95% Confidential Interval	Upper Bound of The 95% Confidential Interval
θ_1	28.705	1.639	24.998	32.413
θ_2	12.921	1.69	9.897	16.744

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